Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/11/2007
PI Name:	Borhan, Ali Ph.D.		
Project Title:	Dynamics of Drops and Bubbles in Confined Flows of Complex Fluids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Co-PI: Dr. H. Haj-Hariri Dept. of Mechanical & Aerospace Engineering University of Virginia Charlottesville, VA 22904		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	04/01/2002	**End Date:**	09/30/2006
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2006, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:			
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We perform an investigation of the motion and deformation of fluid particles in a tube, where the drop and/or the suspending fluid are non-Newtonian. The proposed research is aimed at exploring the non-linear effects of bulk rheology on the underlying behavior of drops and bubbles, and distinguishing the resulting features from those in Newtonian fluids.
The experimental portion of the work considers the motion of fluid particles through transparent tubes, using a variety of Newtonian and non-Newtonian immiscible fluid pairs. The effects of the confining geometry on the mobility and deformation of the fluid particle are examined over a wide range of the dimensionless parameters (such as Reynolds and capillary numbers, the ratio of drop size to tube radius, etc.). The critical values of the dimensionless parameters that lead to transitions in drop shape (e.g., pointed and knife-edge cusps, unsteady shapes, tip-streaming, breakup) and migration speed are also identified. The theoretical portion of the proposed work focuses on the development and implementation of a computational algorithm for the numerical simulation of the problem. The computational approach is based on a hybrid algorithm that combines the best features of the Volume-of-Fluid (VOF) method for two-phase flow and the level-set technique which is capable of accurately representing the topological features of fluid-fluid interfaces. The numerical predictions for the mobility and shape of the drop are compared to the corresponding experimental measurements.

Task Description:

The main objective of the proposed research is to address some of the outstanding issues involving fluid particle motion in viscoelastic two-phase systems. In particular, we propose to examine the role of fluid particle-confining boundary interactions in the dynamics of fluid particles in such systems by performing a comprehensive experimental and theoretical/computational investigation of the confined motion of fluid particles in cylindrical and periodically-constricted tubes. Our goal is to perform the following tasks: 1. Generate quantitative and qualitative data on the behavior of fluid particles for comparison with the computational results. 2. Develop a simulation tool for treating the kinematics, the nonlinear constitutive equations, and the interfacial conditions. 3. Develop simple models that describe each basic mechanism underlying our experimental and computational observations.

The results of the proposed study will lead to improved insight into the role of non-Newtonian fluid behavior in the dynamics of fluid particles in confined domains, and will further our understanding of complex interfacial flows. The extension of the volume-of-fluid and level-set methods to the case of non-Newtonian flows involving fluid-fluid interfaces in the proposed computations will also represent a significant outcome of the proposed research, with considerable potential in other applications involving interfacial flows which are so frequently encountered under microgravity conditions.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

During the past two years, we accomplished the following tasks:
1. We completed a parameter study of the motion of Newtonian liquid drops through a tube at moderate Reynolds number, using the code we had developed earlier based on the VOF/level set algorithm. One graduate student was involved with this computational effort.

2. We developed a code based on the finite volume method to examine drop motion in a tube for the case in which one of the fluid phases was viscoelastic. The FENE-CR constitutive equations were adopted for modeling the stress terms, and the artificial compressibility factor was introduced in the continuity equation to effectively increase the convergence rate of the computations. A boundary-fitted grid conforming to the deforming bubble shape was generated at each time step by solving the requisite elliptic partial differential equations. We found the formation of a protrusion at the trailing end of the drop when the external phase was viscoelastic, and a flattening of the trailing end when the internal phase was viscoelastic. One graduate student was involved with this computational effort.

3. Under certain conditions (e.g. high Deborah and capillary numbers), a sharp axisymmetric tail at the trailing end of a bubble moving in a viscoelastic fluid developed into a three-dimensional cusp with a "knife-edge" shape. We performed a temporal three-dimensional stability analysis of the axisymmetric base shape to elucidate the underlying physics of this phenomenon. We found the mode with azimuthal wavenumer n = 2 to be the only unstable mode. We investigated the energy budget of the disturbances to determine the production and dissipation terms affecting the growth of the three-dimensional instability. One graduate student was involved with this computational effort.

4. We completed our analytical/computational study of the buoyancy-driven motion of viscoelastic drops within an unbounded Newtonian fluid domain. In the weakly elastic limit, we used domain perturbation to obtain analytical results for the steady shape and migration velocity of the drop when the rheological behavior of the drop phase was modeled using the Oldroyd-B or FENE-CR constitutive equations. The drop was found to deform into an oblate spheroid, with the resulting shape deformations tending to increase the drag on the drop and elastic effects having the opposite effect. For moderately elastic drops, we developed a code based on a combination of the boundary integral and finite volume methods to numerically determine the steady drop shape and mobility. One graduate student was involved with this computational effort.

5. We used the hybrid boundary integral/finite volume code to numerically examine the stability of the shape of translating viscoelastic drops to axisymmetric initial perturbations. For large initial deformations, the drop does not reach a steady shape, and drop breakup occurs through either the formation of an elongated tail, or the development of a re-entrant cavity at the rear. These mechanisms of drop breakup are similar to those reported earlier for Newtonian drops. The effects of elasticity on the critical conditions for drop breakup are currently being examined. One graduate student was involved with this computational effort.

6. We continued our experimental investigation of the effect of surfactants on drop breakup in buoyancy- and pressure-driven motion through tubes. Two undergraduate students were involved with this experimental effort.

Bibliography Type: Description: (Last Updated: 01/16/2007)

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